

Algorithmic Economics: Aggregating Preferences

Sumit Goel

August 26, 2026

Design problem

- An economic environment has three key ingredients:
 - ① a set of agents,
 - ② a set of possible outcomes,
 - ③ agents' preferences over these outcomes.
- An *institution*, or *rule*, determines how an outcome is selected.
- Goal: design rules that satisfy desirable properties.

- **Voting:** voters have preferences over candidates; what voting rules lead to desirable outcomes?
- **Auctions:** an auctioneer is selling an object to multiple bidders; what auction formats lead to desirable outcomes?
- **Allocation:** several siblings are dividing an inheritance; what allocation rule leads to a desirable division?
- **Matching:** students and schools have preferences over each other; what matching rules lead to desirable matchings?

This module

- We take an abstract view of the design problem, without focusing on any particular application.
- We develop a framework to formally describe and study such problems.
- We introduce two desirable properties:
 - ① efficiency,
 - ② incentives.
- We discuss which rules satisfy these properties.

- Let X denote the set of *alternatives*: the possible outcomes of the design problem.
- In our examples:
 - 1 **Voting:** X is the set of candidates.
 - 2 **Auctions:** X contains the possible allocations of the object and payments.
 - 3 **Allocation:** X is the set of possible divisions of the inheritance.
 - 4 **Matching:** X is the set of possible matchings between students and schools.

Preference relation

- A *preference relation* on X is a binary relation

$$\succsim \subseteq X \times X$$

that is:

- ① *complete*: for every $x, y \in X$,

$$x \succsim y \quad \text{or} \quad y \succsim x \quad \text{or both;}$$

- ② *transitive*: for every $x, y, z \in X$,

$$x \succsim y \quad \text{and} \quad y \succsim z \quad \implies \quad x \succsim z.$$

- The statement

$$x \succsim y$$

means that the agent regards x as at least as desirable as y .

Strict preference and indifference

Given a preference relation $\succsim$ on X :

- we say that x is *strictly preferred* to y , written $x \succ y$, if

$$x \succsim y \quad \text{and} \quad y \not\succsim x.$$

- we say that x and y are *indifferent*, written $x \sim y$, if

$$x \succsim y \quad \text{and} \quad y \succsim x.$$

- $N = \{1, \dots, n\}$: set of agents.
- X : finite set of alternatives, with $|X| = K$.
- Each agent $i \in N$ has a strict preference ordering $\succ_i$ over X .
- $\mathcal{P}$: set of strict preference orderings over X .
- A *preference profile* is a collection

$$(\succ_1, \dots, \succ_n) \in \mathcal{P}^n.$$

- A *social choice rule* is a function

$$f : \mathcal{P}^n \rightarrow X.$$

- For each preference profile $(\succ_1, \dots, \succ_n)$, the rule selects an outcome

$$f(\succ_1, \dots, \succ_n) \in X.$$

- Goal: identify good social choice rules—rules that select outcomes that appropriately reflect agents' preferences.

Examples of social choice rules

- *Plurality rule*: an alternative receives a point whenever it is ranked first by an agent; the alternative with the highest total score is selected, with ties broken according to a fixed rule.
- *Scoring rules*: fix scores

$$s_1 \geq s_2 \geq \cdots \geq s_K.$$

An alternative receives s_k points whenever it is ranked in position k ; the alternative with the highest total score is selected.

- *Dictatorship*: fix an agent i and always select their most-preferred alternative.
- *Constant rule*: fix an alternative $x \in X$ and select x for every preference profile.

Illustration: Different rules

Consider the following preference profile:

$\succ_1, \succ_2$	$\succ_3, \succ_4, \succ_5$	$\succ_6, \succ_7$
<i>a</i>	<i>a</i>	<i>b</i>
<i>b</i>	<i>b</i>	<i>c</i>
<i>c</i>	<i>d</i>	<i>d</i>
<i>d</i>	<i>c</i>	<i>a</i>

- Plurality selects *a*.
- The Borda rule, with scores (3, 2, 1, 0), selects *b*.
- Thus, different social choice rules may select different outcomes at the same preference profile.

Desirable property: Unanimity

- A social choice rule f is *unanimous* if, whenever all agents most prefer the same alternative x ,

$$f(\succ_1, \dots, \succ_n) = x.$$

- Unanimity captures a minimal notion of **efficiency**: if everyone agrees on the best outcome, the rule should select it.
- Which of the earlier examples are unanimous?

Desirable property: Strategyproofness

- A social choice rule f is *strategyproof* if, for every agent i , every $\succsim_i \in \mathcal{P}$, every $\succsim_{-i} \in \mathcal{P}^{n-1}$, and every possible misreport $\succsim'_i \in \mathcal{P}$,

$$f(\succsim'_i, \succsim_{-i}) \not\succsim_i f(\succsim_i, \succsim_{-i}).$$

- That is, no agent can obtain a strictly better outcome by misreporting their preference.
- Strategyproofness captures **incentives**: agents' preferences are often their private information, and the rule should not give them a reason to misreport. Otherwise, strategic behavior may lead to unintended outcomes.
- Which of the earlier examples are strategyproof?

Theorem 1.

Suppose $|X| \geq 3$. A social choice rule is unanimous and strategyproof if and only if it is a dictatorship.

There is no non-dictatorial rule that is both **unanimous** and **strategyproof**.

Summary

- We studied rules for aggregating individual preferences over alternatives into a societal outcome.
- The impossibility result demonstrates that there is no single rule that would work in all environments.